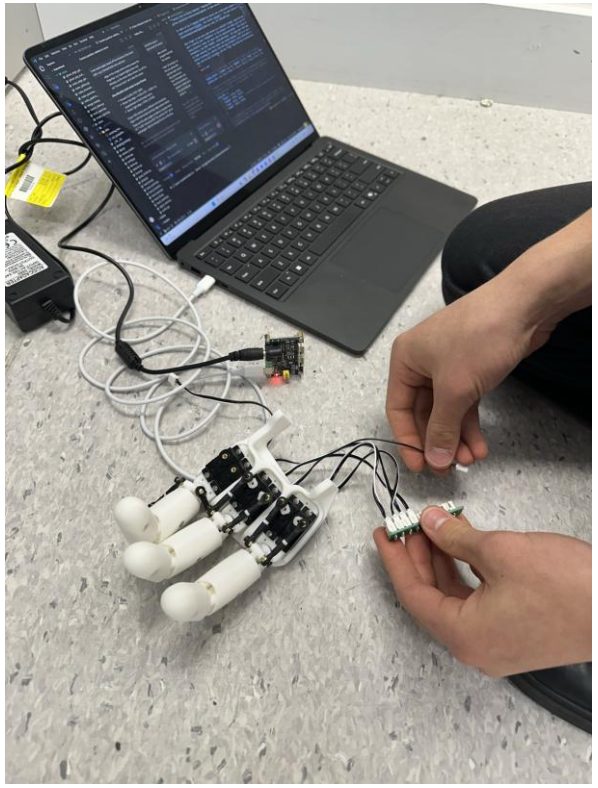
VISION-GUIDED ROBOTIC HAND

Design brochure

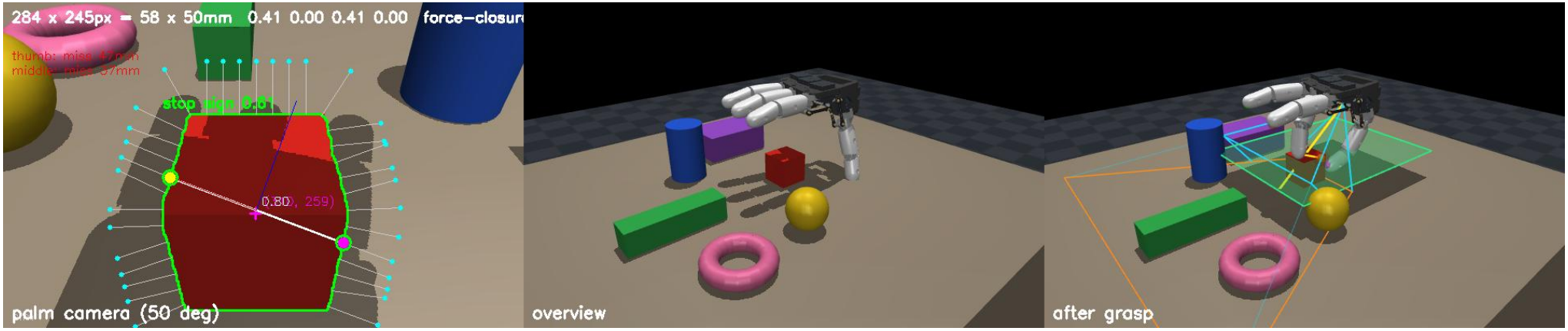
DESIGN GOAL

Design a robotic hand that uses one ordinary camera to plan where each finger should touch an object, with no grasp-training data and prove the plan in simulation before moving the real hand.

Ethan Bayley and Brandon Lai



Our hand in real life.



Our simulation. Left: the palm camera finds the cube and outlines it. Centre: overview. Right: the hand with the grasp plane (green) and camera view (orange and blue lines).

CONTENTS

01 - CONTENTS

1 Significance (pages 3-5)

03 The problem, goal and constraints
04 Who it helps and why it matters
05 Existing work

2 Methodological approaches (pages 6-11)

06 Criteria and schedule, planned and actual
07 Materials and technologies and why
08 Risk assessment
09 What we planned to build
10 Three options
11 How the design developed

3 Creativity and innovation (pages 12-19)

12 Mind map
13 Our design in sketches
14 The final design and its size
15 The simulation and why these tools
16 Main stages and what is new
17 Speed optimisation
18 Outlines and inner contours
19 Grip points and force vectors

4 Execution, construction and testing (pages 20-34)

20 Grasp plane, camera and kinematics
21 Steady grasp and object permanence
22 Results and evaluation (simulation)
23 How we built the real hand
24 Servos, finger order and the camera mount
25 Moving the camera
26 The main app and the servo tuner
27 Testing the real hand
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30 Every test in one place
31 Criteria for the hand and apps, evaluated
32 How far we met our aims
33 Taking it further: the next version
34 Skills, decisions, reflection and next steps

5 Presentation and communication

The whole brochure, including the cover

Credits

35 Credits and sources

02 - KEY EVENTS, IN ORDER

1 - Planned our own hand
A finger and a 16-motor board drawn, not finished

2 - Saw the scope was too big
Pivoted and weighed three options

3 - Chose our own edge detection
Tested on Pollen's hand in simulation first

4 - Outline found
YOLOE mask plus a hole search: 87% match

5 - Fingertips made steadier
423 down to 193 px/s

6 - Hidden edges filled in
93% found when the object stays still

7 - Picking up tested
80 of 80 lifted in the physics simulation

8 - Built the real hand
8 of 8 servos pass our bus test

9 - Moved the camera
Palm view lost the object as the fingers closed

PROBLEM

03 - THE PROBLEM

People who use a prosthetic hand usually pick each grip themselves. They tense muscles in the arm to switch between a few preset hand shapes, which takes training, is slow and often gives a grip that does not suit the object. Many people stop using their prosthesis. Studies put the number anywhere from 9% to 80% and a lack of useful function is a common reason. Hands that work well are also expensive.

It is meant for people who use a powered prosthetic hand. Everyday life means holding many different objects and choosing each grip by hand is slow and tiring. A hand that could look at an object and plan where each finger goes would take that work away. That is what we set out to build: find the object's edge, plan where each finger touches and test the grip in simulation.

9% to 80%
of people stop using their prosthesis: 9% in a US-Japan study, 30 to 80% in a narrative review

US$9k to $40k
what a myoelectric hand cost in one case study (US$9,000 to $40,000)

4 of 5
our proposed target: simulated grips that succeed, for each object on a plain background

04 - OUR ENGINEERING GOAL

Write a vision pipeline that plans valid grasp positions from one 2D image of an object, with at least 4 out of 5 successful simulated grips per object on a plain background.

It breaks into three parts, each tested on its own: find the object's edge, plan where each finger touches and test the grip in simulation (pages 16 to 22). Then the same code drives the real hand (pages 23 to 29).

It is mainly a software goal. We use Pollen's open-source hand, so the new work is the planning, the camera mount and the control apps (page 10).

Six criteria turn this into tests (page 6); each is scored on page 22. Who it helps and why it matters: page 4.

Proposed target. How far we met it: page 32.

05 - CONSTRAINTS WE WORKED WITHIN

- **Time:** about two weeks of planned work before 23 October (page 6).
- **Hardware:** our own hand was not finished and Pollen's hand arrived late in the project.
- **Camera:** one small monochrome camera, with no depth.
- **Computer:** a laptop CPU, no graphics card.
- **Safety:** nothing goes to real motors until it has passed a dry run.

Sources: US-Japan prosthesis rejection study (PMC6571339); narrative review (Prosthesis 2025, 7, 167); myoelectric hand cost case study (PMC5698676). Full references on page 35.

IMPACT

06 - WHAT CHANGES FOR THE PERSON USING THE HAND

	Today: a powered hand with preset grips	With our planner
Choosing a grip	Tense arm muscles in a learned pattern to step through a few preset hand shapes	Point the hand at the object; the camera finds it and the planner places each fingertip
Fit to the object	A preset shape may not suit this object	Contacts are planned on this object's own outline, holes included, like a mug handle (page 18)
A new object	The same few presets	No grasp-training data: any object the detector can name (about 4,500 words)
Before it moves	The hand closes when told to	Forces are checked to cancel before any finger moves (page 19)

The aim: take the work of choosing a grip away from the user, so the hand is quicker and less tiring to use.

07 - WHO IT HELPS

- **People with a powered prosthetic hand,** the main users: fewer muscle patterns to learn.
- **People who fit and train prosthetic hands:** a hand that plans its own grip could take less time to learn.
- **Makers and researchers:** an open-source hand, one ordinary camera and a laptop CPU, with no depth camera or graphics card.

08 - WHY IT MATTERS TO SOCIETY

- **Many people stop using their hand** (9% to 80% in the studies on page 3); a lack of useful function is a common reason. A hand that is easier to use could help more people keep using theirs.
- **Cost:** hands can cost US$9,000 to $40,000. Our planner runs on parts that cost far less: a small camera and a laptop.

09 - WHY IT MATTERS TO ENGINEERING

- **Every fingertip is placed,** where the systems on page 5 choose one grip type or one point.
- **No training data:** a new object needs no new photos.
- **Checks before moving:** force vectors must cancel first.

Not a medical device. Our results come from simulation and a bench hand, not from people using it.

Our comparison, from pages 3 and 5 and our own results. The 9% to 80% and cost figures are from the sources on page 3.

EXISTING WORK

10 - WHAT OTHER GROUPS DID

Other groups have tried to let a prosthetic hand choose its grip from a camera. All four papers below train a neural network on labelled photos and two add depth hardware. They work on a fixed set of objects, pick a grip type or one grasp point and never place each fingertip. **Ours needs no grasp training and places every fingertip.**

System	Trained on labelled photos	What it chooses	Extra hardware
Castro et al. (2022)	Yes	One of five hand postures	Palm camera, Raspberry Pi
Wang et al. (2022)	Yes	A grasp type	None built
Zhang et al. (2025)	Yes, 22 objects	One grasp point	Depth camera
Stracquadanio et al. (2025)	Yes	Grasps from a trained generator	RTX 3080 graphics card
Ours	No	A contact point for every fingertip	One camera and a laptop CPU

11 - CASTRO ET AL. (2022)

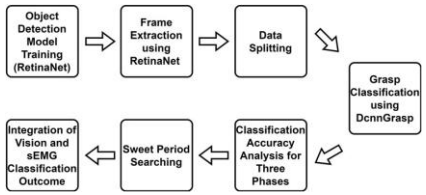


A 3D-printed arm with a palm camera; a trained network names one of five hand postures.

Where it fell short

- Trained on 25,000 to 39,000 labelled photos
- Tripod was called pinch 41 times out of 42

12 - WANG ET AL. (2022)

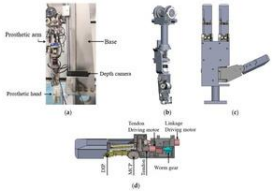


A network names the grasp type from the first 320 ms of a reach. No hand was built.

Where it fell short

- Vision plus muscle signals (90.06%) scored below vision alone (91.59%)
- Never says where a finger goes

13 - ZHANG ET AL. (2025)

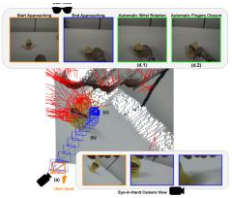


A depth camera and YOLOv3-tiny, trained on 22 objects; a symmetry axis picks one grasp point.

Where it fell short

- Needs a depth camera and a trained detector
- No success rate over many trials

14 - STRACQUADANIO ET AL. (2025)



Guesses depth from one photo; a trained generator proposes grasps (RTX 3080).

Where it fell short

- Mugs failed most (success 0.58)
- Only 3 amputee volunteers

Figures: Castro 2022 (CC BY), Wang 2022 (CC BY), Zhang 2025 (CC BY 4.0), Stracquadanio 2025 (arXiv, shown for review with credit). Full references on page 35.

PLANNING

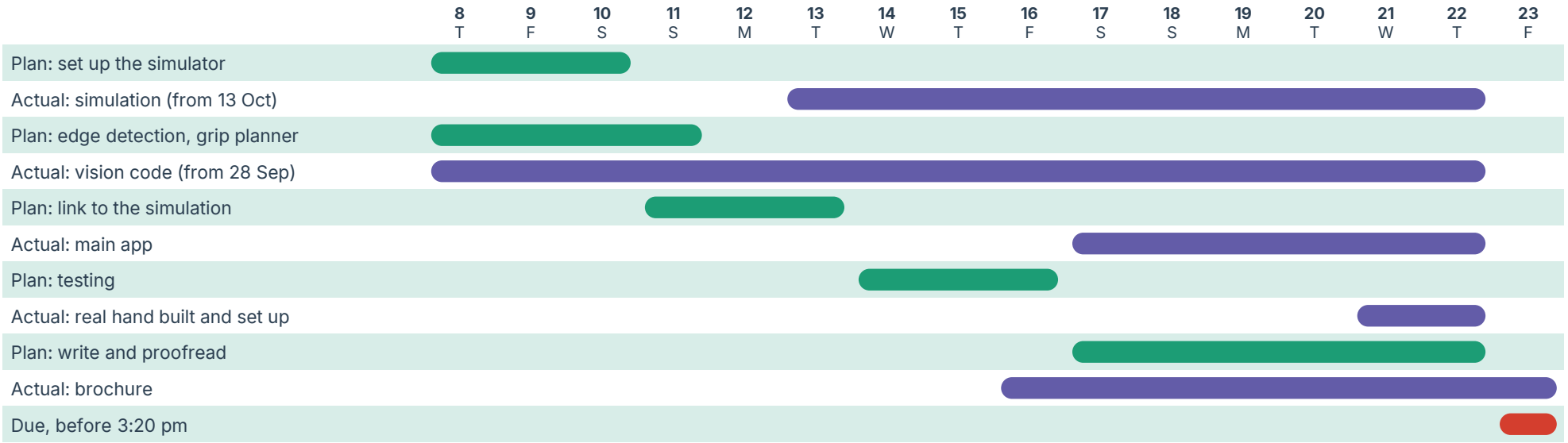
15 - SUCCESS CRITERIA

Criterion	Proposed target
1 Find the outline	4 of 5 trials per object, plain background
2 Valid grip plan	4 of 5 trials per object
3 Reach the target	Fingertip within 3 mm
4 Hold the object	Held 5 s, 4 of 5 trials
5 Unseen objects	4 of 5 trials per unseen object
6 Run fast enough	A loop time of 200 ms or less

We set the targets before testing. Each one can be checked with a count or a measurement, so the verdicts on page 22 are not opinions.

Why these numbers: 4 of 5 lets one failure through in a five-trial test, so one unlucky trial does not fail a method that works. 3 mm is small next to a fingertip pad, so a contact that close still lands on the planned spot. 200 ms gives about five plans a second, fast enough for the plan to keep up while the hand reaches.

16 - OUR SCHEDULE: PLANNED (GREEN) AND ACTUAL (PURPLE)



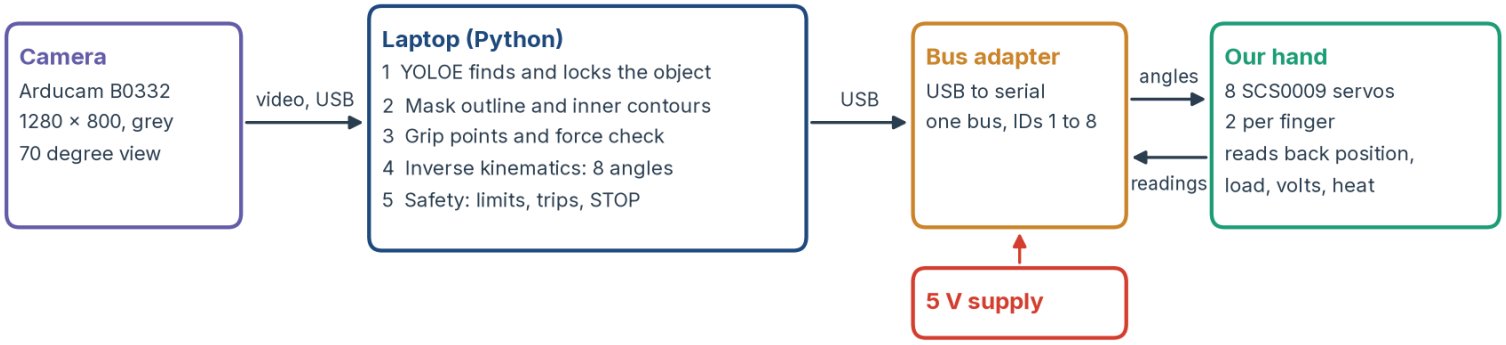
Planned dates: our plan. Actual: when the work first appears in our project history; the vision code was started before this plan was written.

MATERIALS

17 - WHAT WE CHOSE, WHAT ELSE WE LOOKED AT AND WHY

Part	Chosen	Also looked at	Why we chose it
Hand	Pollen's Amazing Hand, open source	Our own hand (page 9)	Already designed and printable; Pollen's MuJoCo model gives the exact finger linkage for the simulation
Printed parts	PLA	None	Cheap and quick to reprint when a part breaks
Motors	8 Feetech SCS0009 servos	Our N10 gearmotors and board	Two fit in each finger; one serial bus; each reports position, speed, load, volts and heat; 2.3 kg.cm at 6 V
Camera	Arducam B0332 (OV9281)	A depth camera; a phone or webcam	Small enough for the hand; global shutter, so no smear while moving; 100 frames a second over USB. We found late that it is grey only
Finding the object	YOLOE	A detector we train; SAM	Names about 4,500 kinds of object with no training on ours; SAM took 1,896 ms an outline
Outline	YOLOE's own mask	Otsu, Canny, GrabCut, SAM	82% match in 3.5 ms; GrabCut scored 87% but about 20 times slower (page 18)
Simulator	MuJoCo	Building the hand first	Real contact and friction, so a grasp can be tested before the hand exists
Computer	A laptop CPU	A small board in the forearm	Fast enough to develop on (104 ms a loop); a board is the next step (page 33)
Code and apps	Python, Tkinter	None	The same planner and hand driver run in the simulation and on the real hand

18 - HOW THE PARTS CONNECT



Servo figures: Feetech SCS0009 listing and Pollen's documentation. Camera: Arducam B0332 product page. Times: our own runs (pages 17 and 18).

RISKS

19 - RISK ASSESSMENT

Risk	Likelihood	Consequence	What we did about it
Simulation differs from real life	High	High: results may not hold on the hand	Labelled every result as simulation; friction assumed 0.5; tested the real hand when it came
The real camera has no depth	High	Medium: contacts off by a height error	Camera-only modes and the grasp plane; measured the error (17 to 18 mm)
Small tests can flatter results	Medium	Medium: results look better than they are	Reported counts and failures, not only successes
Hands arrive late	High (happened)	High: no time to test on hardware	Wrote the real-hand driver in advance; built and set up the hand on 21-22 October
Fingers pinch or crush	Medium	High: injury or broken parts	Dry run first; one finger at a time; saved limits; a trip on load, heat or lag
Weak power supply	Happened	Medium: servos miss commands	Retries and a low-voltage readout; a stronger supply is the fix (4.7 V measured under load)
Servo overheats	Low	Medium: a damaged servo	Trip at 55 degrees C (our estimate)
Wrong finger order	Happened	Medium: the wrong finger moves	Found on the hand; renamed in the code; tests check it
Camera loses the object as the fingers close	Happened	High: the plan is blind when it matters most	Moved the camera to the side of the hand (page 25); to be tested

Our judgement: likelihood and consequence are our own ratings; "happened" marks risks that came true.

FIRST PLAN

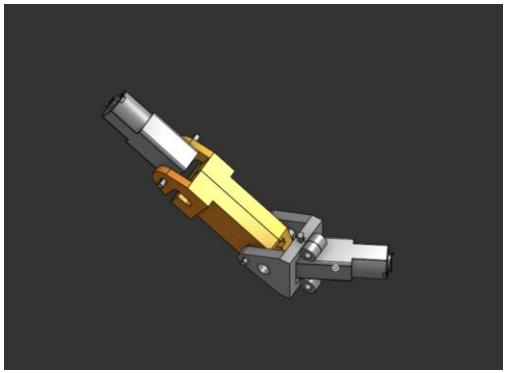
20 - OUR FIRST PLAN

Our first plan was to build the whole hand ourselves. Each finger joint would be driven directly by a small N10 motor, with bevel gears turning the finger. Direct drive has no long cables or gearbox between motor and finger, so the joint can be pushed backwards by hand (it is back-drivable). The current each motor draws also tells us how hard that finger is pushing, so we get a torque reading from every joint without a separate sensor. We aimed for a hand under US$200 with the motors, sensing and controller built in.

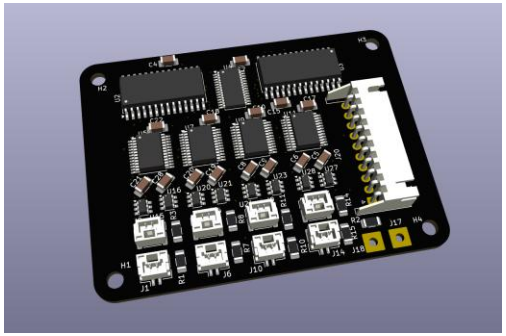
We drew a finger driven by two small DC gearmotors through a linkage (top right). We also drew a custom board to control all 16 motors for position control: eight motor drivers, a chip that makes the PWM signals, two chips that set directions and a current sensor on every channel (bottom right).

Then we added it up. A whole hand means printing and assembling a thumb and three fingers, building and tuning the board, writing the control code and then writing the vision software that decides where the fingers go. **We realised the scope was far bigger than the time we had, so we had to pivot.** The next page shows the options we came up with.

21 - OUR FINGER AND MOTOR BOARD



Our finger (CAD)



Our motor board (KiCad)

Started, not finished. Neither was built or tested.

22 - THE PLAN IN FOUR PARTS

Part of the plan	What it means
Direct drive	No gearbox, so the joint is back-drivable
Bevel gears	Small N10 motors turn the finger joints
Motor board	16 motors, position control, current sensing
Low cost	An aim of under US$200, all hardware built in

These are aims, not results.

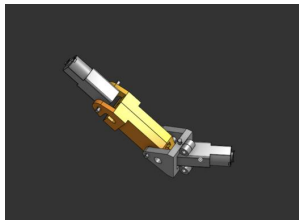
Images: our own CAD and KiCad screenshots.

OPTIONS

23 - CUTTING THE WORK DOWN

When we saw the scope on page 9, we came up with three options that each cut down what we had to do and weighed them. Our own copies of Pollen's hand had not arrived yet, so any option that needed hardware had to wait for them.

24 - A FINISH OUR OWN HAND



Our finger CAD

Positives

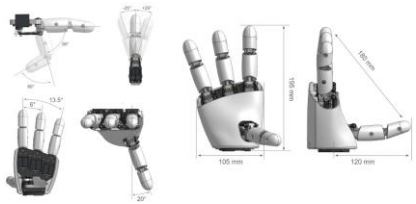
- We choose the motors and finger shape
- The board could read motor current for grip force

Negatives

- Needs printing, building, wiring and tuning
- Not enough time

Not chosen. Building, wiring and tuning a whole hand, then writing the vision code, would take far longer than we had (page 9).

25 - B OPEN-SOURCE HAND, PRETRAINED GRIPS



Pollen's Amazing Hand: 4 fingers, 8 motors, 195 mm tall

Positives

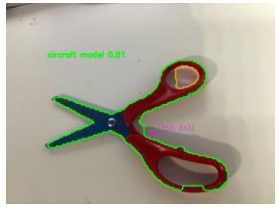
- A proven hand to build on
- A trained network picks a grip type

Negatives

- Needs labelled photos for every object
- It only knows the objects and grips it was trained on

Not chosen. It needs labelled photos of every object and only knows what it was trained on, the limit the papers on page 5 hit.

26 - C OPEN-SOURCE HAND, ADAPTIVE CAMERA FEEDBACK



Our edge detection: outline and holes

Positives

- No training data: our own edge detection works on any picture
- Test at once in simulation on the open-source hand

Negatives

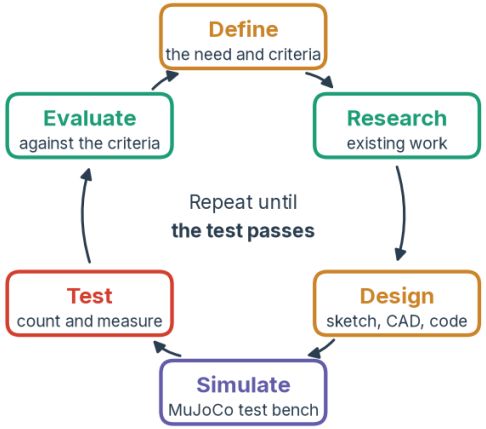
- Results are simulation only until the hands arrive
- Friction and camera effects are assumed

Chosen. No training data, works on any picture and testable at once in simulation on Pollen's hand.

Images: our own CAD; Pollen Robotics, AmazingHand repository (Apache 2.0 code; mechanical design CC BY 4.0); our pipeline on a test photograph (not our own).

DEVELOPMENT

27 - OUR DESIGN CYCLE



Every change on the right went round this loop. A test showed a problem, we changed one thing and then we measured again, so each step can be checked.

One change at a time, so we always knew which change caused the result.

28 - EACH CHANGE AND THE TEST THAT LED TO IT

What a test showed	What we changed	Result after	Page
Building our own hand would take longer than we had	Used Pollen's open hand; wrote the software first	Planner tested in simulation before the hands arrived	9-10
One frame took 2.4 s; the SAM outline was 1,896 ms	Outline from YOLOE's mask; crop around the object	104 ms a loop	17
The mask missed holes such as a mug handle	A search for inner contours	Match 82% to 89%	18
Fingertips placed on the outline lifted nothing	Force vectors must point in and cancel	Forces checked before moving	19
Palm-down grasps: nothing got under the object	Grasp plane tipped 40 degrees	80 of 80 lifted in simulation	20
Our camera is grey only	Kept the mask outline, the method that coped best in grey	87% match in grey	20
Fingertips jittered on live video	The plan changes only when a new one is clearly better	423 to 193 px/s	21
A closing finger hid part of the object	Object permanence: redraw the hidden edge	Hidden edge found 38% to 93%	21
Servos failed the first bus test	Move to the middle first, then spin	8 of 8 within 0.6 degrees	27
The palm camera lost the object as the fingers closed	Moved the camera to the side of the hand	Design change, to be tested	25

Each row is a change recorded in our design logs, with the test before and after it.

MIND MAP

29 - MIND MAP OF OUR PROJECT

Vision-guided robotic hand

Problem
- Grip chosen by tensing muscles: slow
- 9-80% stop using their hand
- Hands cost US$9k-40k
- A preset grip may not fit the object

Options
- A our own hand: too big
- B pretrained grips: needs data
- C adaptive camera feedback: chosen

Constraints
- About two weeks
- One grey camera, no depth
- A laptop CPU
- The hand arrived late
- Nothing moves before a dry run

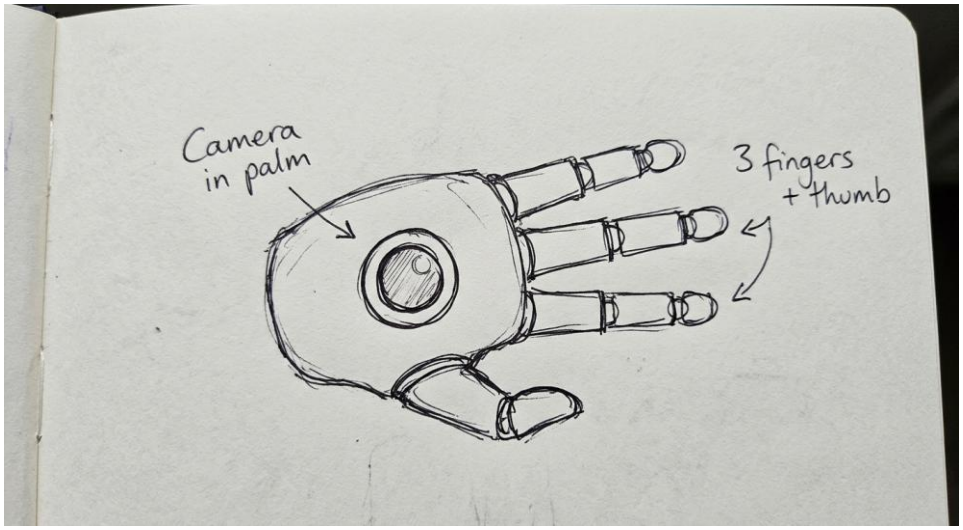
Tests
- Simulation pick-up 80 of 80
- Outline match 87%
- Hidden edge 38% to 93%
- Servo bus 8 of 8

Ideas
- Grasp plane
- Force vectors that cancel
- Object permanence
- Steadiness rule
- Camera at the side

Mind map drawn on computer. Problem figures: page 3.

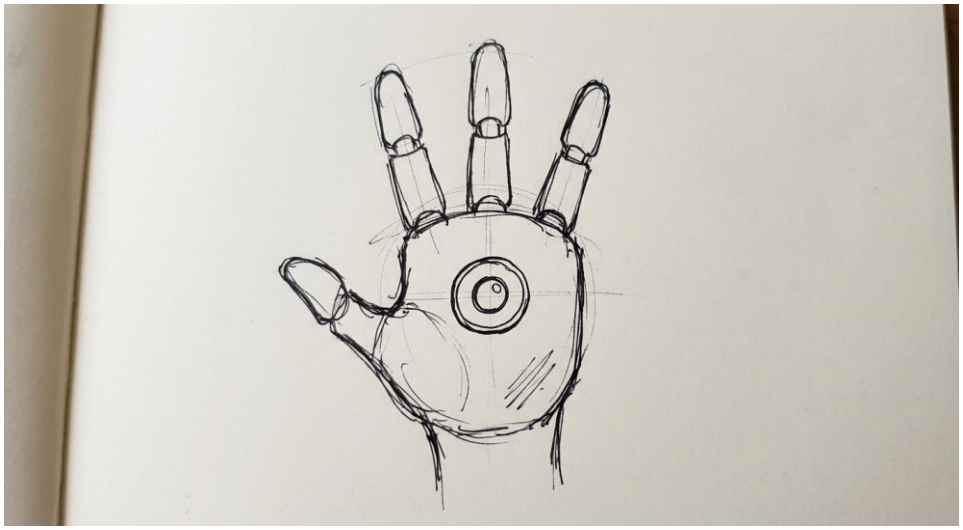
SKETCHES

30 - OUR DESIGN IDEA IN SKETCHES



Camera in the palm

Three fingers and a thumb; the camera looks out of the palm



The palm from the front

The camera sits in the middle of the palm

31 - WHAT THE SKETCHES SETTLED

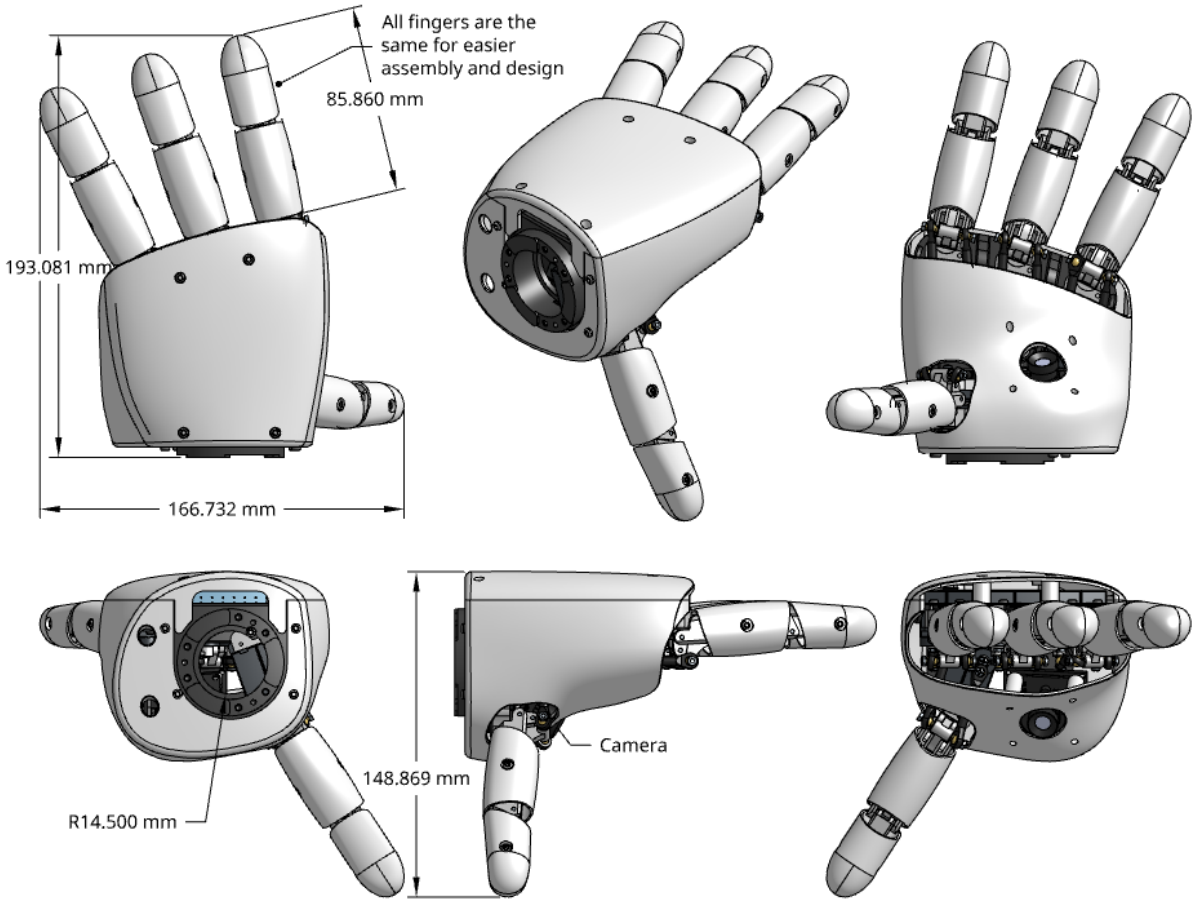
- Use Pollen's open hand: two servos for each of the four digits.
- Add one camera in the palm, tipped towards the fingertips, so it sees what the fingers are about to grasp and moves with the hand.
- One serial bus carries all eight servos to the laptop, where the code from the simulation plans the grasp.

Our sketches came first: the camera in the palm and three fingers plus a thumb became the CAD design on page 14.

Sketches: our own pen sketches. Not to scale.

FINAL DESIGN

32 - OUR HAND WITH THE PALM CAMERA



Six views of the final design, with its main dimensions.

33 - KEY DIMENSIONS

166.7 mm
wide

193.1 mm
tall

148.9 mm
deep

85.9 mm
finger length

34 - WHAT THE DESIGN SHOWS

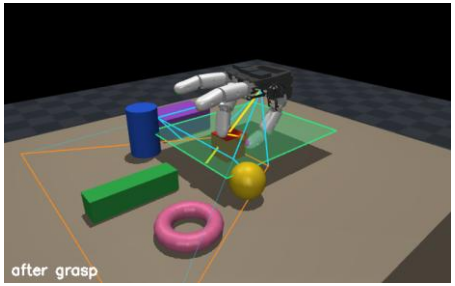
- **All four digits use the same finger,** for easier assembly and design.
- **The camera sits under the palm,** looking past the fingers at the object.
- **A round wrist mount** (14.5 mm radius) at the back holds the hand.
- It follows our sketches (page 13) and Pollen's open design.
- Why each part and material was chosen: page 7.

Later change: the camera moved to the side of the hand (page 25).

Drawing: our CAD of the hand with the camera mount, based on Pollen Robotics' Amazing Hand (CC BY 4.0). Dimensions are from the CAD, not measured on the built hand.

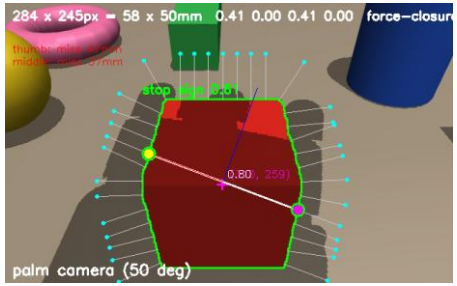
SIMULATION

35 - 3D VIEW



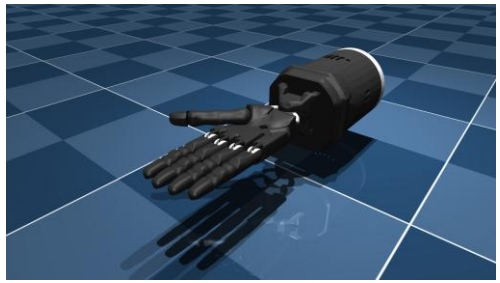
The grasp plane and the camera's field of view.

36 - PALM CAMERA VIEW



The YOLO outline (green) and fingertip contacts (cyan).

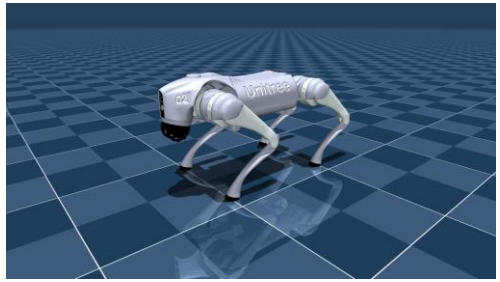
37 - OTHER SIMULATIONS



Shadow Hand



Franka Panda arm



Unitree Go2 robot

38 - WHY WE SIMULATED

We did not have enough time to build and test a hand and our Pollen hands had not arrived. We wanted to write the code first and know it works, so we built the test bench on a computer. The simulation helped a lot. We tuned where each fingertip goes and which fingers can reach, checked that the force vectors cancel before anything moved and, because it runs real contact and friction, saw whether the hand would grip an object and lift it. When the hands arrived we built one and moved the same code across (page 23).

39 - SET-UP AND WHY THESE TOOLS

MuJoCo 3.11.0, Python 3.13.14, OpenCV 4.13.0.92, Ultralytics 8.4.161. We added collision shapes, a palm camera and our own inverse kinematics to Pollen's model.

Tool	Why we used it
MuJoCo	Pollen's own MuJoCo model of the hand gives the exact finger linkage; it simulates contact and friction
Pollen's hand	Open source and already designed: 4 fingers, 8 servos; we built ours when it arrived (page 23)
YOLOE	Names objects from about 4,500 words without training on ours
OpenCV	Ready-made Otsu, Canny and GrabCut outlines to compare (page 18)
Arducam B0332	Small USB camera, global shutter, 100 frames a second; monochrome, as we learned late (page 20)

Hand model: Pollen Robotics (Apache 2.0, made from their Onshape CAD). Other simulations: MuJoCo Menagerie, Google DeepMind (Apache 2.0). Screenshots: our own simulation.

PIPELINE

40 - MAIN STAGES OF THE PIPELINE



Camera image

One flat frame from the hand's camera (here a test photo). No depth and no object name yet.



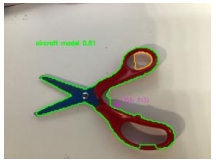
YOLOE finds the object

YOLOE boxes each object it can name, from about 4,500 words. We lock onto one box (green).



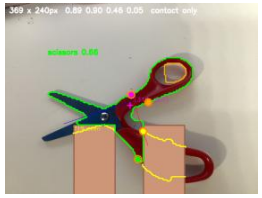
Only the box's pixels

Everything outside the box is dropped, so the next steps work on a small picture and run faster.



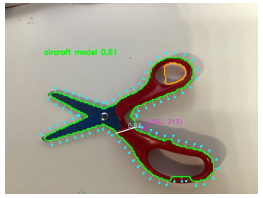
Edge detection

YOLOE's mask gives the outer contour (green); a hole search finds inner contours (orange).



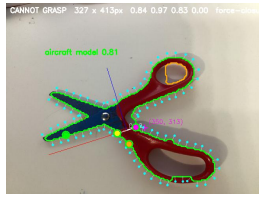
Object permanence

A remembered snapshot of the outline redraws any edge a finger covers (yellow).



Grip candidates

A point every 8 pixels round the contour, moved out by a fingertip's thickness. Cyan: a finger can reach.



Finger placement

The best pair whose pushes cancel becomes thumb and finger contacts; IK turns them into servo angles.

Each arrow: the output of one stage is the input of the next.

41 - THE FINAL VERSION

The final version locks onto one object and checks only a small square around it, with a full scan twice a second. It remembers the object's shape so a covering finger does not change the plan. A detection takes about a tenth of a second, so in between it slides the outline along with the object using optical flow, which tracks how pixels moved from one frame to the next. It then plans each fingertip, runs inverse kinematics to get the two motor angles for each finger and holds the grasp steady. It takes the outline straight from YOLOE's mask (page 18).

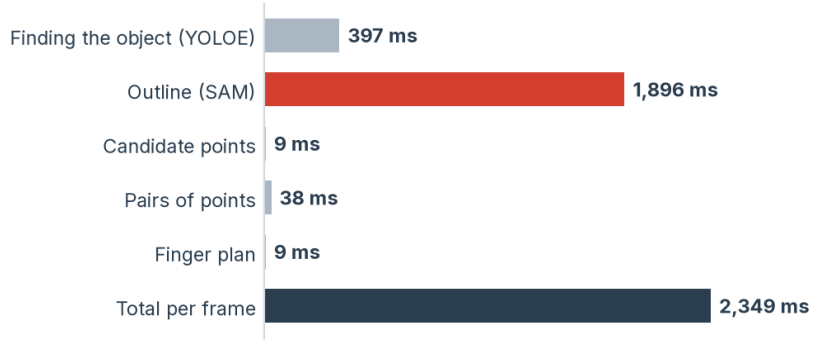
42 - WHAT IS NEW IN OUR SOLUTION

What we came up with	Page
Lock on one object	21
YOLOE in its own process	17
Mask outline, 20 times faster	18
Inner contours to grip	18
Force vectors at the centre	19
Grasp plane at a natural angle	20
Object permanence	21
Camera at the side of the hand	25

Images: our pipeline on a test photograph (not our own).

SPEED OPTIMISATION

43 - WHERE THE TIME WENT AT THE START



Median per stage with the SAM outline, on a busy laptop. Total: the five stages added up.

44 - MAKING IT FASTER

When we first ran the whole pipeline end to end, one frame took about 2.4 seconds (2,349 ms) with the SAM outline (about 1.1 seconds with GrabCut). That is far too slow to follow a moving object. Timing each stage showed that two of them took almost all of it: finding the object with YOLOE (397 ms) and drawing the outline (1,896 ms). Grip planning took under 60 ms.

We fixed the slow stages one at a time: a faster outline, a smaller picture, looking only at a crop around the locked object and moving YOLO into its own process. **The finished loop takes 104 ms** (median, about 10 frames a second), inside our target of 200 ms (page 6).

45 - THE BIGGEST COSTS, BEFORE AND AFTER



What it cost: smaller GrabCut crops fit worse (84% to 79% match), a 480 px picture finds small objects less reliably, a crop can miss a new object for up to half a second and the mask outline sometimes marks highlights as holes. Not tested on the single-board computer.

The charts come from different test runs on one laptop, so the totals are a guide, not one measurement.

OUTLINES

46 - FIVE WAYS TO OUTLINE THE MUG

We tested five ways to outline an object inside YOLOE's box, on the same mug: green is the outline, orange a hole (the handle should be one).



Otsu

27 ms: Found the handle but called the logo a hole.



Canny

1 ms: Outlined only the logo.



GrabCut, fast

51 ms: Good outline, a little blocky.



GrabCut, slow

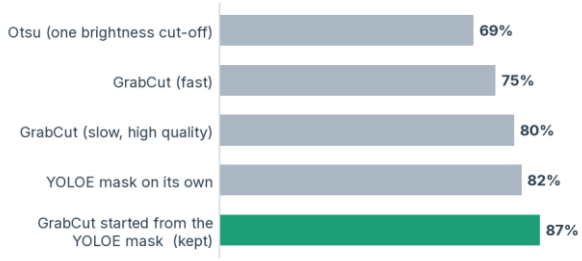
130 ms: Closer fit, but a false slot in the rim.



SAM

639 ms: Closest, but far too slow.

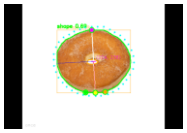
47 - MATCH WITH THE REFERENCE OUTLINE



Match with SAM's outline over 8 photos (100% is exact). Green is the best score.

48 - WHAT WE KEPT AND WHY OUTER AND INNER CONTOURS

GrabCut started from YOLOE's mask scored highest (87%), but the mask alone scored 82% about 20 times faster, so we kept the mask. **A search for holes lifted it to 89%** on 10 objects in colour (87% in grey). It still fails when the detector boxes the wrong thing.



Doughnut



Scissors



A false one

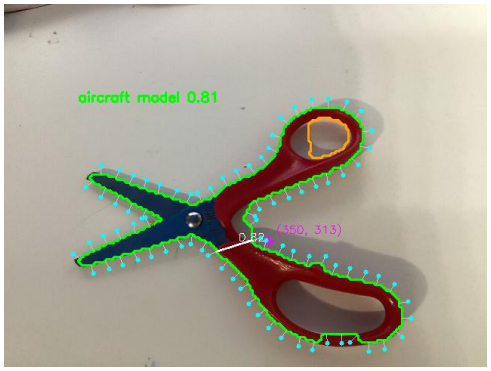
Why both: the outer contour is the object's silhouette, where fingertips squeeze from outside. Many objects have holes, like a mug handle or scissor loops. With only the outer contour the planner treats them as solid and may put a fingertip in empty space or join two contacts across a gap. The inner contour marks where the material really is, so a grip pair never crosses a hole and a finger can push outward inside a loop, the way people hold scissors.

A hole counts only if it is enclosed, at least 0.5% of the object and 30 mm across, so highlights do not make false ones.

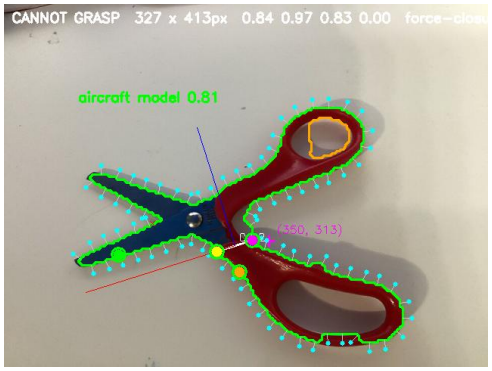
Pictures: our pipeline on test photographs (not our own). Reference outline from SAM; no hand-labelled outlines.

GRIPPING

49 - FROM OUTLINE TO CONTACTS



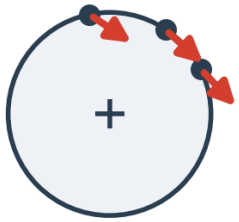
Points on the contour



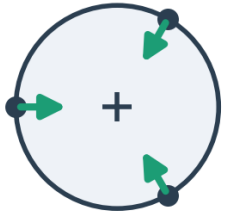
Chosen contacts

50 - WHY THE DIRECTION OF PUSH MATTERS

Any positions: it slips



Pushing at the centre: it holds



An illustration, not data. Arrows are the push of each fingertip.

51 - GRIP POINTS

We place a point every 8 pixels around the whole contour. Each is pushed out from the edge by the thickness of a fingertip, measured on the grasp plane, so it marks where the fingertip pad would sit. Then we work out which points a finger can reach: cyan dots can, red crosses cannot.

From the reachable points we choose pairs. The line between the two points must stay inside the object, so it never crosses a hole and each push must lie inside its friction cone so the fingertip grips instead of sliding. Narrower pairs near the centre of mass score higher.

52 - FORCE VECTORS

In the simulation, moving fingertips to points along an object and closing the hand lifted nothing. The fingers slipped off, or the object skidded away. What matters is the force each fingertip applies.

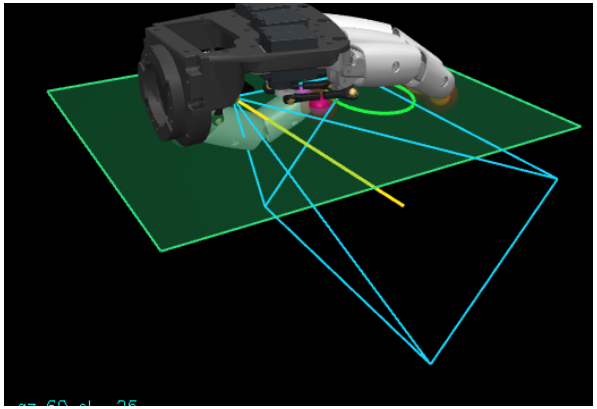
So for every contact we work out a force vector, the direction and size of its push and accept a set of contacts only when the vectors point at the centre of the object and cancel out. With balanced forces the fingers squeeze the object and it stays put.

Balance alone is not enough: in one test all 5 grasps balanced but only 4 were real pinches, because not every finger reached the object.

Pictures: our pipeline on a test photograph (not our own). Friction cone after Nguyen 1988 (friction 0.5 assumed).

GRASP PLANE

53 - THE GRASP PLANE

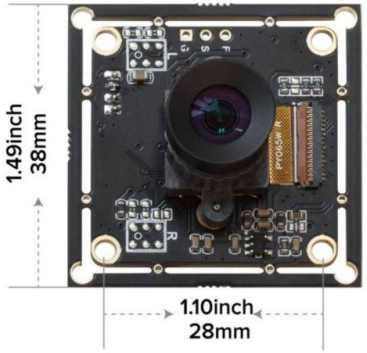


Grasp plane (green)

A camera gives a flat picture and the fingers need 3D positions. We did not know where to put the fingertips, so we made a grasp plane: a flat sheet in front of the hand that we assume the object's top lies on. Each contact point becomes a ray from the camera and where the ray meets the plane is where the fingertip goes.

Tuning it taught us it must line up with the hand. Palm straight down, the thumb squeezed upward and nothing got under the object. **Tipped 40 degrees, the way a person reaches, the fingers closed from the side and lifted 80 of 80 objects.** A height error h puts a contact 0.58 h to the side.

54 - THE CAMERA

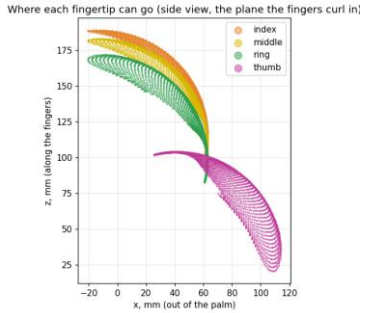


Arducam B0332

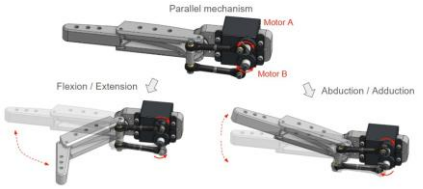
Our camera (Arducam B0332) is monochrome. We found that out late. Without colour, YOLOE has less to tell objects apart and the edge detector loses the colour difference. Our first grey test dropped the outline match from 80% to 57% and a dark mug on a dark table fell from 96% to 8%. The mask method coped best, at 87%.

The view is 70 by 47 degrees, so the thumb's reach sits at the left edge and objects there were cut off. One camera cannot measure distance, which is why we need the plane. Where the camera sits: pages 24 and 25.

55 - INVERSE KINEMATICS



Fingertip reach



Pollen's finger

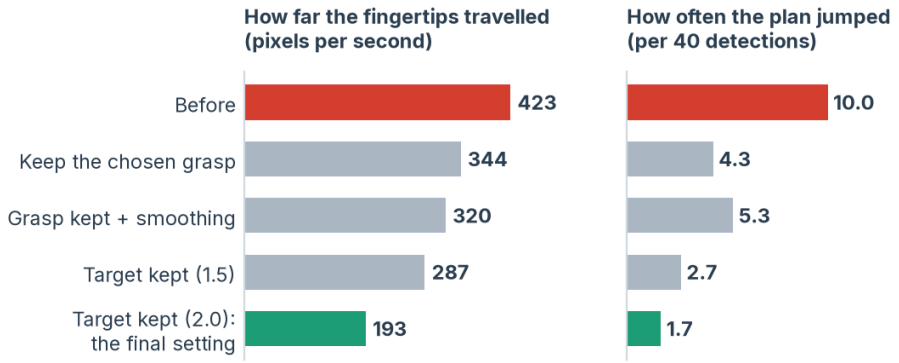
We know where a fingertip should be and need the two servo angles that put it there. We sampled Pollen's linkage every 10 degrees; the table matches Pollen's model to 0.17 mm on average.

Each finger has two motors. We kept that design because it is simple and it is the one Pollen gave us. Turning both the same way swings the finger sideways; opposite ways curl it.

Kinematics from Pollen's model, not measured on a real hand. Pollen finger picture: AmazingHand repository (Apache 2.0, CC BY 4.0). Camera: Arducam B0332 product photo.

STEADINESS

56 - KEEPING THE GRASP STEADY

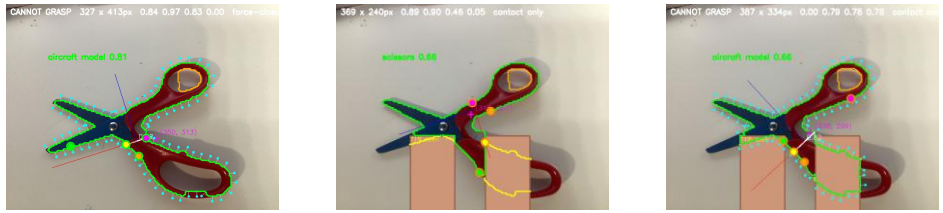


Each bar is one setting. Red is where we started, green is what we kept. Lower is steadier.

On live video the fingertips jittered. Every detection gave a slightly different outline, so the grasp positions moved a little each time and the fingers moved with them. We tried smoothing the positions and a rule that the chosen target and grasp only change when a new one is clearly better. Smoothing helped little; the rule did most of the work.

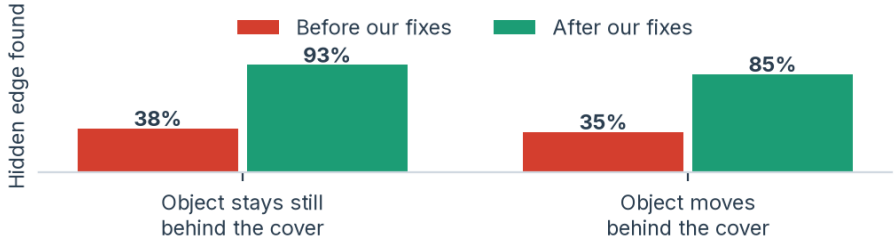
We kept all of it. Fingertip travel fell from 423 to 193 pixels per second and the plan jumped 1.7 times instead of 10 in 40 detections. The cost: about 0.6 seconds of delay when the object really changes.

57 - OBJECT PERMANENCE



1 Seen clearly → **2 Covered** → **3 Edge filled in**

Two bars stand in for fingers; yellow is the remembered outline. When the fingers close on an object they cover part of it, so the detector sees a smaller shape and the plan changes at the moment the fingers should hold still. So we gave the planner object permanence: it takes a snapshot while the object is clear, finds where it matches what is still visible and redraws the hidden edge.



Steadiness: 3 photos, 40 detections each, noise added. Hidden edge: 7 photos, 168 painted covers per bar, not real fingers; not run on a live camera.

SIMULATION EVALUATION

58 - RESULTS AGAINST OUR CRITERIA

Criterion (page 6)	What we measured in simulation	Verdict
1 Outline	87% match with the reference; wrong box on 1 of 8 photos	PARTLY MET
2 Grip plan	5 of 5 plans balanced (aimed camera); 4 of 5 were real pinches	MET
3 Reach the target	10 of 10 fingertip aims within 3 mm	MET
4 Hold the object	Lifted 80 of 80 with a steady hand; 59 of 80 with a shake	PARTLY MET
5 Unseen objects	5 unseen simulated objects; 4 of 5 grasped	MET
6 Speed	Loop about 104 ms on a laptop; target 200 ms	MET

The physical hand: we built and tested the real hand after these simulation results; see pages 23 to 31. Hold the object is only partly met: with a shake added, 59 of 80 is below 4 of 5.

59 - PICKING UP OBJECTS



Physics pick-up test: 5 small objects, 16 random starts each; success is lifted 50 mm.

60 - BENEFITS

- No grasp-training data
- About 100 ms on a laptop CPU
- Forces checked before moving
- One ordinary camera
- Holes found: mug handle 4 of 4, doughnut and bagel 2 of 2 (page 18)
- Steadier fingertips: 423 to 193 px/s (page 21)

61 - LIMITATIONS

- Simulated camera has depth
- Friction (0.5) is assumed
- Small samples (5 to 10 trials)
- Real-hand results: page 31
- Grey camera: outline match fell from 80% to 57% in our first test (page 20)
- Hidden edges tested with painted covers, not real fingers (page 21)

Samples are small; these results are from the simulation.

BUILD PROCESS

62 - THE BUILD, IN THE ORDER WE DID IT

Our Pollen hand arrived after we had written and tested the software in simulation. We laid every part out, built one finger first as Pollen's guide suggests and checked it moved smoothly, then built the others and the whole hand, wired the eight servos to one bus and set each servo up in our tuner (page 24).

63 - OUR PHOTOS



1 Parts laid out

Every part on the desk first



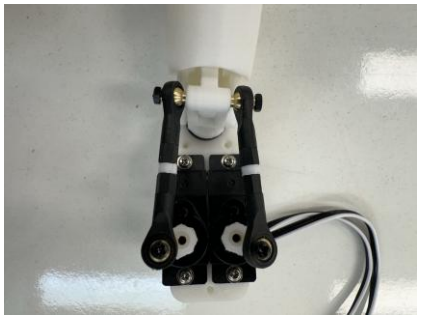
2 Preparing parts

Sanding and fitting printed parts



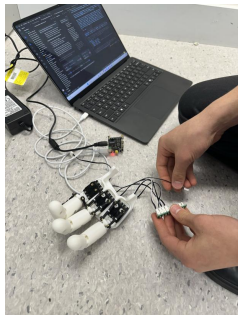
3 One finger

One finger on its SCS0009 servo



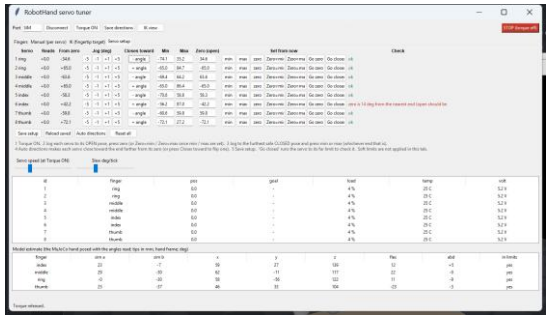
4 The servo pair

Two servos and the linkage rods



5 The whole hand

Wired to the bus and the laptop



6 Servo setup

Zero, range and direction in our tuner

Photos 1 to 5: our own. Picture 6: a screenshot of our servo tuner.

SETTING UP

64 - SERVOS: ZERO, RANGE AND DIRECTION

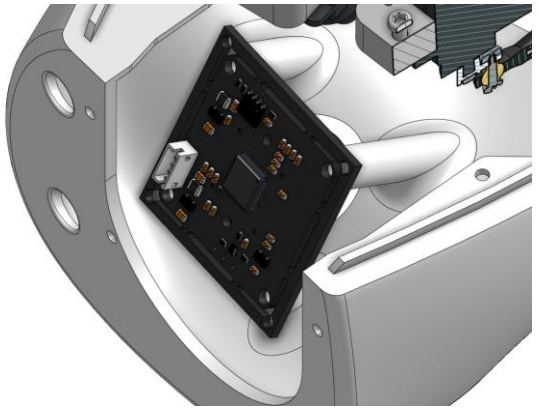
Step	What happened
1 Servo IDs	IDs 1 to 8 were already set. Servos 7 and 8 first did not answer: a power and adapter problem, now gone
2 Zero position	All eight moved to the middle (0 degrees) and spun 30 degrees each way and back: all within 0.6 degrees
3 Range	Each servo's zero, minimum, maximum and closing direction saved in a settings file; every program keeps to them
4 Direction	Servos that closed the wrong way were reversed in the setup table
5 Finger order	On our hand the index is servos 5 and 6 and the ring is 1 and 2, the reverse of the guide; we swapped the names in the code
6 Thumb	The real thumb needed 68% of its travel to touch the index, where the simulation needs 44%, so its travel is scaled by 1.55 (one measurement)

65 - THE FIRST CAMERA MOUNT: IN THE PALM

Pollen's guide has no camera, so the mount is our own. We measured its base: 20.7 mm left, 26.6 mm forward and 20.5 mm down from the centre of the round mount at the back of the hand, pointing 30 degrees down. The planner uses these numbers. We later moved the camera (page 25).



The camera in the palm (CAD) from outside



The camera in the palm (CAD) from inside

Camera mount measured by us on the real hand. CAD: our own.

CAMERA REDESIGN

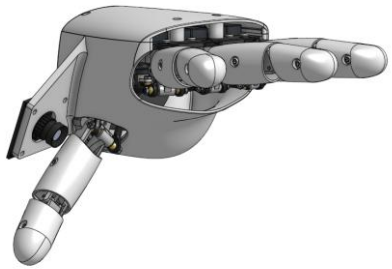
66 - WHY THE PALM CAMERA DID NOT WORK



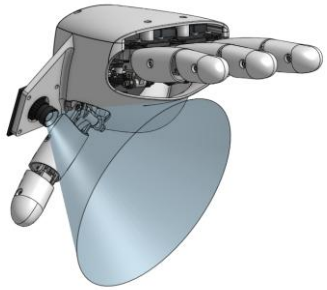
Before: the lens looks out of the palm.

As the hand reached in and the fingers began to close, the object came very close and the fingers came between it and the lens. The camera could not see the object, or the parts we wanted the fingertips on, at the moment the plan mattered most.

67 - THE NEW POSITION: OUTSIDE THE HAND, BETWEEN INDEX AND THUMB



After: the camera on the outside



Its field of view (blue cone)

We moved the camera to the outside of the hand and tipped it to look between the index finger and the thumb. That is where the fingertips meet the object, so the grip area should stay in view while the fingers close.

68 - BEFORE AND AFTER

	Palm camera (first)	Side camera (now)
Where it sits	Under the palm, looking out	Outside the hand, between index and thumb
While the fingers close	Fingers and the close object block the view	Looks into the gap the fingers close on
Mount measured	20.7, 26.6, 20.5 mm; 30 degrees down	Not yet
Tested on the hand	Mount used by the planner	Not yet

69 - WHAT HAPPENS NEXT

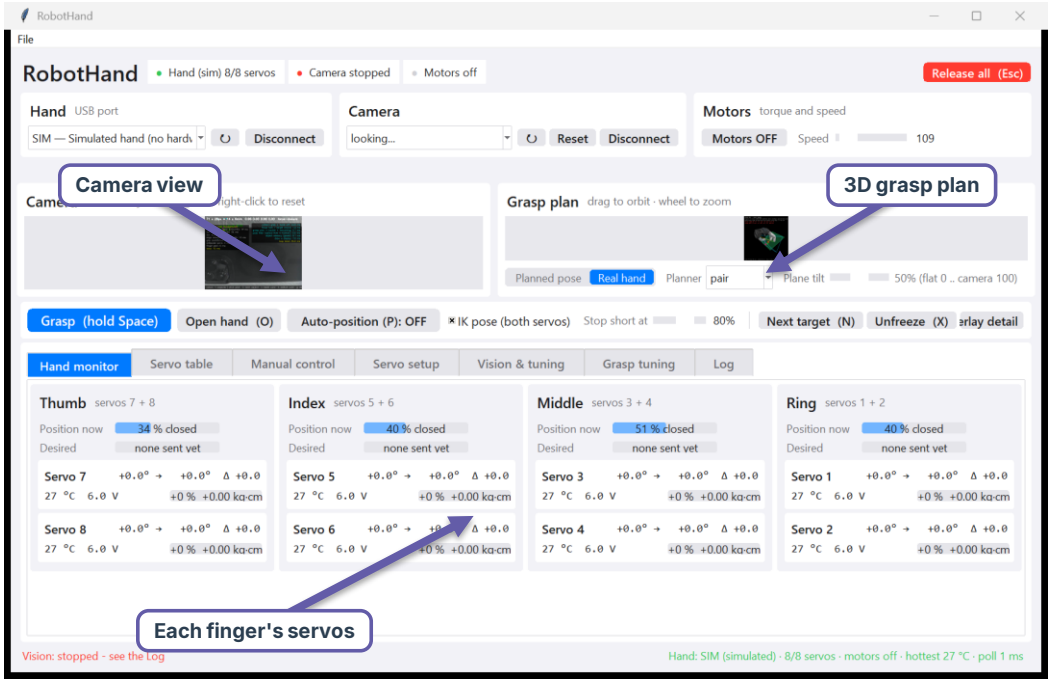
1. Fit the new mount; measure its position and angle.
2. Enter them in the planner.
3. Check the object stays in view while closing.
4. Run the grasp test (page 28).

Design change only. Not yet tested on the hand.

CAD: our own, based on Pollen Robotics' Amazing Hand (CC BY 4.0). The blue cone is the camera's field of view drawn in CAD.

OUR APPS

70 - THE MAIN APP

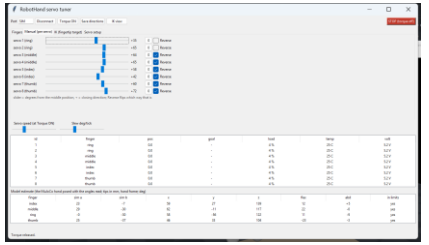


Our main app: camera, grasp plan, motors and every servo's readings.

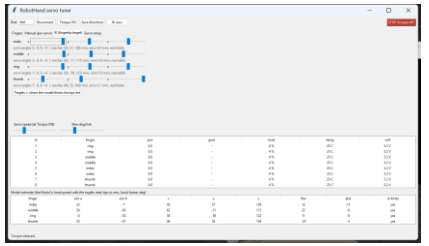
Tabs hold the servo table, manual control, servo setup, vision and grasp tuning and a log. Written in Python (Tkinter), so the camera code, planner and hand driver are the same as in the simulation.

Screenshot with the simulated hand: the readings are not real; it shows the layout.

71 - THE SERVO TUNER



Manual: a slider for each servo



Fingertip targets through our IK

72 - SAFETY BUILT IN

- Connecting only reads; nothing moves until motors are switched on.
- The first goal is where each servo already is, so the hand cannot jump.
- Every goal stays inside the saved limits.
- Release all (Esc) and a red STOP turn torque off.
- Our tests never move the motors.

Screenshots: our own apps.

TESTING

73 - SERVO BUS TEST ON THE REAL HAND

The test checks that every ID answers, moves all 8 servos to the middle, spins each one 30 degrees each way and back and passes a servo only if it ends where it was sent.

8 of 8
servos answered, IDs 1 to 8

0.6 deg
largest error after the spins and at the middle

5.1-5.3 V
supply voltage at rest

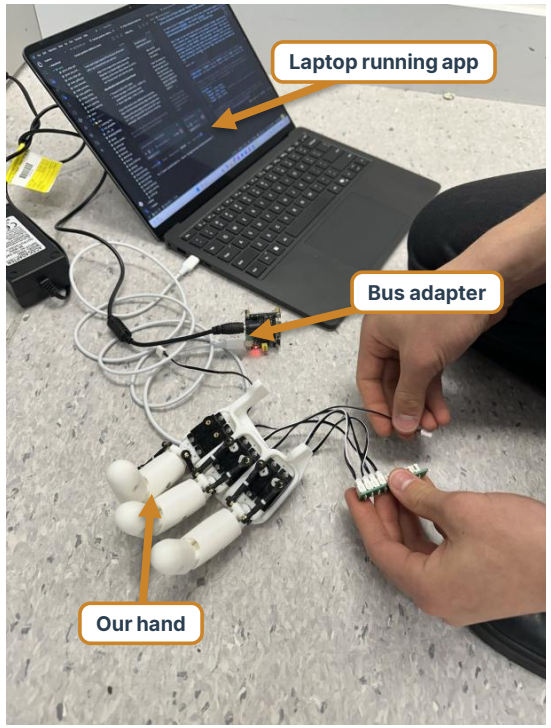
4.7 V
supply when the fingers moved

All pass: every servo answered, spun and came back to the middle within 0.6 degrees.

74 - WHAT WENT WRONG AND THE FIX

- **False fails.** Our first test spun each servo around its start. Servos 3 to 8 started near the end of their travel and failed by up to 317 degrees (the angle wrapped round). Fix: move to the middle first, then spin.
- **Servo 6 timeouts.** Servo 6 sometimes missed a command. The supply fell from 5.2 V to 4.7 V when the fingers moved, below the servo's 4.8 V minimum. The app now retries and shows the low voltage; the real fix is a stronger supply.
- **Wrong finger order.** The index and ring were swapped against the guide; we renamed them in the code.

75 - THE TEST SET-UP



The hand on the servo bus, wired to the laptop.

Next: grasps on real objects, planned on page 28.

Measured on the real hand, 22 October 2026. Photo: our own.

GRASP TEST PLAN

76 - WHAT THE TEST CHECKS

Hand criterion 6 (page 31): the real hand lifts an everyday object 4 times out of 5. It is the first test of the whole chain on hardware: camera, plan, real fingers.

The pass rule is the one we used in simulation (criterion 4, page 6), so the two results can be compared directly.

Pass: held for 5 s in at least 4 of 5 tries, for each object.

77 - OBJECTS AND CONDITIONS

- **Five everyday objects** of different shapes: one with a hole (like a mug), a round one, a box, a bottle and a soft one.
- **The same marked spot** in front of the hand for every try, on a plain table.
- **5 tries per object:** 25 grasps in all.
- **Camera:** in the position on page 25, once its mount is measured.
- **Also recorded:** which fingers touched, the planning time and the supply voltage.

78 - PROCEDURE, EVERY TRY

1. Place the object on the marked spot.
2. Run the planner with the motors off; check the planned contacts on screen (dry run).
3. Switch the motors on and close the hand.
4. Lift the hand 50 mm and hold for 5 seconds.
5. Record held, slipped or missed; release and place the object again.

79 - SAFETY DURING THE TEST

- Saved limits on every servo; goals outside them are refused.
- One person on the STOP button (Esc) for every try.
- Hands and loose objects clear of the fingers while they close.
- Stop if a servo trips on load, heat or low voltage (page 27).

Our own test plan. Results go on the next page.

GRASP TEST RESULTS

80 - RESULTS: HELD FOR 5 SECONDS (TICK) OR NOT (CROSS)

Object	Try 1	Try 2	Try 3	Try 4	Try 5	Held (of 5)	Pass?
Total						of 25	

81 - COMPARED WITH THE SIMULATION

	Simulation	Real hand
Lifted and held	80 of 80 steady; 59 of 80 shaken (page 22)	of 25
Objects passed	5 of 5	of 5

82 - WHAT WE NOTICED

- Fingers that touched:
- Failures and why:
- Supply voltage while closing:

Filled in by hand from our test sheet.

TEST SUMMARY

83 - EVERY TEST, ITS CONDITIONS AND THE RESULT

Test	Conditions and trials	Result	Page
Outline methods	5 methods; 8 photos; match against SAM's outline	Best 87% (GrabCut from the mask); mask alone 82%	18
Colour and grey	10 objects, each in colour and grey	89% colour, 87% grey (mask plus hole search)	18
Holes	Mug, scissors, doughnut, bagel	Mug handle 4 of 4; loops all; doughnut and bagel 2 of 2	18
Speed	One recording on a laptop	Loop 104 ms median (target 200 ms)	17
Grip plan	5 plans (camera aimed at the object)	5 of 5 balanced; 4 of 5 real pinches	19
Fingertip reach	10 fingertip aims	10 of 10 within 3 mm	22
Unseen objects	5 simulated objects not used in tuning	4 of 5 grasped	22
Steadiness	3 photos, 40 detections each, 5 settings	423 to 193 px/s; plan jumps 10 to 1.7	21
Hidden edge	7 photos; 168 painted covers per bar; still and moving	38% to 93% still; 35% to 85% moving	21
Pick-up (physics)	5 objects, 16 random starts each; steady and shaken	80 of 80 steady; 59 of 80 shaken	22
Servo bus (real hand)	8 servos; spin 30 degrees each way	All within 0.6 degrees	27
Supply (real hand)	At rest and with fingers moving	5.2 V falling to 4.7 V	27
Grasps (real hand)	5 everyday objects, 5 tries each (planned)	NOT TESTED yet	29

Multiple conditions: colour and grey, still and moving, steady and shaken, simulated and real. Every result is from our own runs.

Fingertip reach, grip plan and unseen objects: our re-tests of 23 October.

HAND EVALUATION

84 - CRITERIA FOR THE HAND AND APPS AND WHAT WE FOUND

Our first criteria (page 6) were for the software. Once we had a hand and apps, we wrote criteria for them too, so each could be tested. The thresholds are our own.

Criterion	How we will know	What we found	Verdict
1 Servos respond	All 8 answer; each within 1 degree of its zero after a move	8 of 8 answer; all within 0.6 degrees	**MET**
2 Safe control	Nothing moves until connected and motors on; goals stay inside the saved limits	Goals clamped and refused when off; covered by tests	**MET**
3 Camera position known	Mount measured and used by the planner	Palm mount measured and used; the new side mount (page 25) not yet	**PARTLY MET**
4 Simulation matches the hand	Closing travel and thumb contact agree, as far as we can judge	Tuned until it looked about right, by eye; thumb from one point	**PARTLY MET**
5 Set up without code	Zero, range and direction set and saved in an app	Done in the tuner and the main app	**MET**
6 Lift a real object	The real hand lifts an everyday object 4 times out of 5	Test planned (page 28)	**NOT TESTED**

85 - WHAT CARRIED OVER FROM THE SIMULATION

- The same planner and kinematics model drive both.
- The simulated servos report the same readings as the real ones, so the same safety checks work on both.
- The habits: dry runs first, one finger at a time.

86 - WHAT IS DIFFERENT ON THE REAL HAND

- Directions, limits and the finger order had to be found on the hand.
- A weak supply: 4.7 V under load.
- The real camera is monochrome with no depth.
- The palm camera lost the object as the fingers closed, so we moved it (page 25).
- No force sensors: a servo's load is motor effort, not fingertip force.

FINAL EVALUATION

87 - DID WE MEET OUR AIMS?

The software in simulation (criteria, page 6)

- **Find the outline: partly met.** 87% match with the reference outline, but the detector boxed the wrong object in 1 of 8 photos.
- **Plan a grip: met.** All 5 plans balanced their forces and 4 of 5 were real pinches.
- **Reach the target: met.** 10 of 10 fingertip aims landed within 3 mm.
- **Hold the object: partly met.** 80 of 80 lifted held steady, but 59 of 80 with a shake added.
- **Unseen objects: met.** 4 of 5 unseen objects grasped. **Speed: met.** 104 ms per loop, inside 200 ms.

The real hand and apps (criteria, page 31)

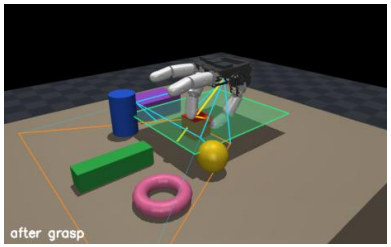
- **Met:** all 8 servos respond, control is safe and setup needs no code.
- **Partly met:** the simulation matches the real hand only by eye; the new camera mount is not measured yet.
- **Not tested:** lifting a real object (test plan on page 28).

Overall

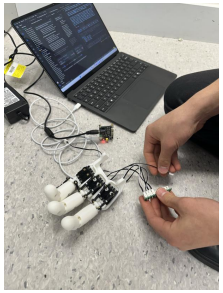
In simulation the planner meets four of its six aims and partly meets the other two. The real hand is built and every servo works; next we must prove it can pick up a real object from the camera's plan.

Biggest limits: results are from simulation with small samples and the real hand has not yet grasped an object.

88 - IN SIMULATION AND IN REAL LIFE



Simulation: the hand grasps the cube



Real life: our hand on the servo bus

89 - MAJOR BENEFITS AND LIMITS

- **Benefits:** no grasp-training data; one ordinary camera and a laptop; forces checked before moving.
- **Limits:** simulated results; small samples; friction assumed; a grey camera with no depth.

90 - FURTHER RESEARCH (PAGE 33)

- Count real grasps and compare them with the simulation.
- Measure the side camera and check the object stays in view.
- Add depth to replace the flat grasp plane.

NEXT VERSION

91 - A COLOUR DEPTH CAMERA WITH A WIDER VIEW

- **Depth** gives each fingertip a real 3D target, so the flat grasp plane is no longer needed.
- **Colour** gives the detector and the outline more to work with: in our first grey test the outline match fell from 80% to 57% (page 20).
- **A wider view** keeps objects in sight near the thumb; at 70 by 47 degrees the thumb's reach sits at the edge of the picture and objects there were cut off.

Why it matters: the camera is the limit we hit most often in testing.

92 - CUSTOM MOVES THE USER TEACHES THE HAND

A user terminal lets each person save the shapes they need every day, beside the automatic grasp. For example, a pointed finger to flip a light switch, a flat hand to push a door or a pinch for a key.

1. Shape the hand with the sliders, or move it by hand with the motors relaxed.
2. Name the shape, for example "light switch" and save it.
3. Call it up with one command or button; the same saved limits keep it safe.

93 - MORE DEGREES OF FREEDOM

- Pollen's hand has 8 servos, so 8 degrees of freedom: two per finger and each finger curls as one piece.
- A human hand has about 21 in the fingers and thumb (27 with the wrist), so each joint can bend on its own.
- 16 or more would let each finger bend at more than one joint, closer to a real hand. Our first board was already drawn for 16 motors (page 9).

Our aim: 16 or more, to mimic a real hand. The cost: more motors, weight and power.

94 - ALL THE COMPUTING ON THE HAND

- Run the camera, detector and planner on a small computer in the forearm, so no laptop is needed.
- Our loop takes 104 ms on a laptop and the detector is most of it (page 17); a board needs a faster detector or an AI accelerator.
- We have not tested the pipeline on a single-board computer yet.

First step: time the pipeline on the board before choosing it.

95 - A BATTERY IN OR ON THE FOREARM

- A battery built into a forearm shell, or a pack strapped to the forearm, so the hand works away from a wall socket.
- It must hold 5 to 6 V at several amps: our supply fell to 4.7 V when the fingers moved, below the servos' 4.8 V minimum (page 27).
- The forearm also houses the computer and wiring, mounted on the hand's round wrist mount.

Design target: a full day of use on one charge (to be measured).

These are proposals for the next version: none of them has been built or tested. Numbers are from pages 20, 17 and 27. Human hand: about 27 degrees of freedom with the wrist, a commonly used count.

REFLECTION

96 - ENGINEERING SKILLS WE USED

Skill	What we did	Page
Programming	Vision pipeline, grasp planner, hand tools; 387 automated tests	16-21
Simulation modelling	Collision shapes, a palm camera and six objects added to Pollen's hand	15
Kinematics	Our IK table matches Pollen's model to 0.17 mm	20
Force analysis	Friction cones and force vectors that must cancel	19
Hardware build	Assembled the hand; wired 8 servos; set IDs, directions, limits	23-27
CAD and electronics	Our finger CAD; a 16-motor board with current sensing	9
Testing	Random starts, painted covers, grey copies, servo bus test	18-22, 27

97 - DECISIONS WE MADE

- Stopped building our own hand and wrote the software first (pages 9-10).
- Chose our own edge detection over a trained detector (page 10).
- Kept YOLOE's mask over GrabCut: about 20 times faster and better on grey (page 18).
- Added force-vector checks when fingertip placement alone lifted nothing (page 19).
- Tipped the grasp plane 40 degrees after palm-down grasps failed (page 20).
- Let the plan change only when a new one is clearly better (page 21).
- Tested every servo on its own before letting the planner move the hand (page 27).
- Moved the camera to the side when the palm view was blocked (page 25).

98 - REFLECTION

We planned to build a hand, drew a finger and a motor board, then saw that building, wiring and tuning a hand would take longer than we had. Pollen's hand was already designed, so we wrote the software first and tested it on a simulated hand. The simulation changed the code more than we expected: it showed that the thumb sits on one side, that balanced forces can still miss the object and that our fingertips shook until the chosen target stopped changing at every detection. Finding that our camera has no colour was a late surprise.

Pollen's hand arrived late. We built it and every servo passed our test (page 27). Putting the camera on the hand showed us that a palm camera cannot see what the fingers are closing on, so we moved it (page 25). The planner's angles are mapped to the real servos, but that mapping has not yet been checked against real fingertip positions and the real camera sees no depth.

99 - NEXT STEPS

1. Fit and measure the side camera mount.
2. Check the angle mapping against real fingertip positions.
3. Repeat the simulated pick-up on the real hand and compare.
4. Trigger each safety trip on purpose to check it.
5. Save the limits inside the servos themselves (their angle-limit settings).

CREDITS

100 - CREDITS AND SOURCES

Hand, software and images

- Pollen Robotics Amazing Hand: hand model and images (Apache 2.0 code; mechanical design CC BY 4.0).
- YOLOE (Ultralytics), OpenCV, NumPy, SciPy, MuJoCo, SAM. Simulation examples: MuJoCo Menagerie, Google DeepMind (Apache 2.0).
- Test photographs of everyday objects are from other photographers (many from Wikimedia Commons; licences in our project files), not our own.
- Our own: build photos, app screenshots, CAD (including both camera positions), board and simulation images.
- Arducam B0332 product photo and camera figures: Arducam product page. SCS0009 servo figures: Feetech, via a retail listing and Pollen's documentation.
- Our own pen sketches; the mind map and diagrams were drawn on computer.

Papers and problem statement

- Castro et al. 2022 (Front Neurorobot 15:751282, Figs 1, 2, CC BY).
- Wang et al. 2022 (Front Robot AI 9:948238, Fig 3, CC BY).
- Zhang et al. 2025 (Biomimetics 10(4):242, Figs 9, 19, CC BY 4.0).
- Stracquadanio et al. 2025 (arXiv:2503.00466, Fig 1 shown for review).
- US-Japan prosthesis rejection study (PMC6571339); narrative review (Prosthesis 2025, 7, 167); myoelectric hand cost case study (PMC5698676).

Our records

- Every number comes from our own project runs, recorded in our design logs as we worked.